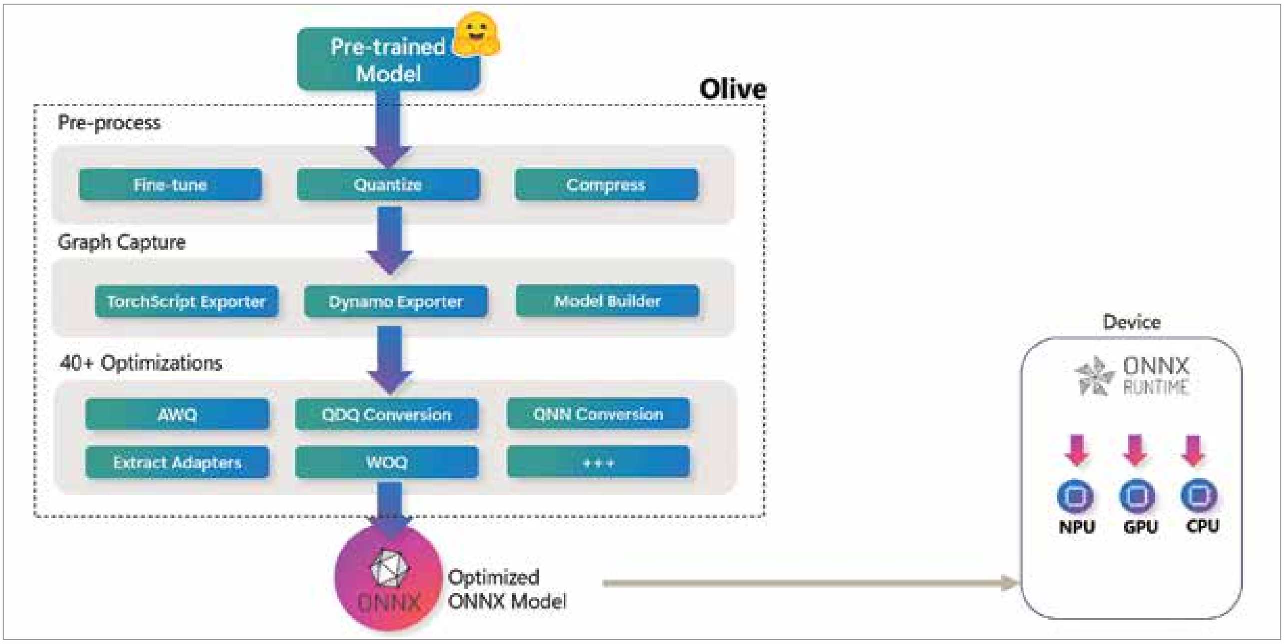

Figure 2: End-to-end ONNX Olive optimisation pipeline

AutoML and low-code AI: Accelerating model development

The AutoML (automated machine learning) frameworks make the whole machine learning process much easier by taking care of the most important tasks like model selection, feature engineering, and hyperparameter optimisation through automation. This leads to a drastic reduction in development time, especially for machine learning applications involving structured and tabular data.

Among the various open source AutoML tools available for Windows platforms, H2O AutoML, Auto-Sklearn, and FLAML are the most popular ones. By using these frameworks, machine learning practitioners with little or no experience in algorithm tuning can also quickly develop strong baseline models.

H2O-based AutoML on Windows: H2O AutoML can automate the entire process of training and evaluating several models and picking the one that performs the best. The following code illustrates this:

AutoML vs low-code AI: A comparison

Aspect	AutoML	Low-code AI
Purpose	Automates model selection, training, and tuning	Simplifies AI development using visual workflows
Coding requirement	Minimal coding or configuration-based	Very low or no coding required
Target users	Data scientists and AI researchers	Domain experts and business users
Customisation level	Limited but performance-oriented	Moderate, UI-driven customisation
Primary use case	High-performing baseline and production models	Rapid prototyping and quick deployment

```
import h2o
from h2o.automl import H2OAutoML
# Initialize H2O
h2o.init()
# Load dataset
data = h2o.import_file("dataset.csv")
# Split data into training and testing sets
train, test = data.split_frame(ratios=[0.8], seed=42)
# Run AutoML
aml = H2OAutoML(max_runtime_secs=300, seed=42)
aml.train(y="target", training_frame=train)
# Retrieve the best-performing model
leader = aml.leader
print("Best model:", leader.model_id)
```

AutoML vs custom models: Choosing the right approach

Automated machine learning (AutoML) and custom models are two different paths to AI development. AutoML is the winner when it comes to speed, accessibility, and quick experiments, while custom models grant more control, flexibility, and